

On / Off

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Abstract

On / Off is an interactive installation composed of 4,000 vintage mechanical switches arranged into tunnel-like modular panels. The piece materializes binary logic as a tactile, performative environment: audience interactions produce emergent combinations of image, light, and sound. Specific switch groups map to conceptual datasets (24 red switches map to ImageNet [Deng et al. 2009]; 266 green switches encode the estimated number of atoms in the universe), while the remainder trigger non-semantic sounds, pointing toward the unknown. The work investigates the tension between deterministic systems and embodied human agency—asking what is revealed when computation becomes touchable.

CCS Concepts

• **Human-centered computing** → *Systems and tools for interaction design*; • **Applied computing** → **Performing arts**.

Keywords

Interactive Light and Sound Installation, Technology and Art, Science and Art, Computational Art, Cybernetic Art, New Media Installation

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1 Concept and Motivation

Contemporary computation often presents itself as invisible infrastructure. On / Off reverses this tendency by rendering binary states legible and material. Each mechanical switch stands for a single bit, and the accumulation of 4,000 such bits becomes a landscape for collective action. The project is motivated by a desire to reconnect algorithmic logic with human tactility, to slow down digital immediacy and reveal the poetics of simple on/off gestures. By mapping limited switch subsets to expansive datasets—ImageNet for image retrieval and an atomic-count surrogate for luminous thresholds—the installation situates perception within a scalar tension between the known and the unknown [Benjamin 2018].

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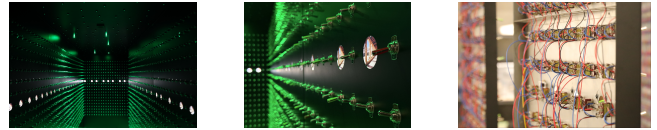


Figure 1: Left: Representative image of the installation ON or OFF — the immersive green-lit environment, sound-light installation. Middle and Right: Detail view of the sound-light installation ON or OFF, showing the modular panel (front and back).

2 System Design and Implementation

On / Off comprises 13 modular panels (each 220 cm × 100 cm), assembled into two tunnel-like structures with a total span of 13 m length × 2.2 m height and a footprint of 13 m × 2.5 m. Each panel integrates: microcontrollers (ESP32) for local I/O, a local image-processing node (Raspberry Pi 4 or NVIDIA Jetson Nano), 80 embedded circular LCDs (2.8" diameter), ceiling LED nodes, and distributed audio playback. Network communication uses Wi-Fi and MQTT; power input is 220 V, stepped down to 24 V and 5 V for modules. Sound is delivered via four active speakers at structural ends; lighting uses addressable LED beads mounted to the ceiling. The software stack supports real-time switch polling, image matching to a pre-filtered ImageNet subset, light control sequences, and non-semantic sound event generation. Modules are fully modular for transport and rapid reconfiguration. Installation requires two technicians and approximately 1.5 days to assemble and test.

3 Interaction and Audience Experience

Participants are invited to flip switches freely; each action updates the state and triggers multimodal feedback. The red switches act as indices into an image database—combinations surface matching imagery on embedded micro-screens—while green switches cumulatively trigger overhead illumination that communicates a conceptual cognitive threshold. Silver switches intentionally remain unassigned to encourage non-semantic sonic events and to preserve open space within the combinatorial field. Typical engagement lasts 3–5 minutes per participant; the system supports multiple simultaneous participants (up to 10), producing both individual and emergent collective behaviors. The installation foregrounds decision, chance, and embodied rhythm: small binary gestures accumulate into visible and audible patterns that challenge notions of control.

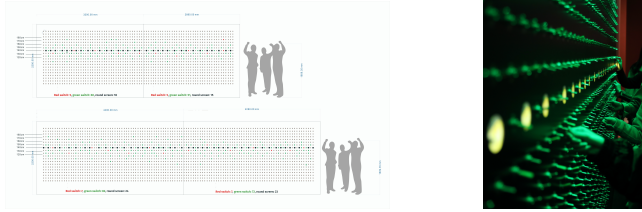


Figure 2: Left: Views of the installation from different angles and scenes of audience interaction. Right: Audience interaction.

4 Contribution and Context

By transforming binary logic into a tactile medium, On / Off contributes to media art practices that interrogate infrastructure, materiality, and embodied computation [Paul 2023]. The work bridges

design, electronic craft, and algorithmic processing, offering a critical and poetic reflection on how digital logics mediate perception. It invites reconsideration of agency within networked systems and proposes physical interfaces as sites for slow, collective reflection on technological limits [Kittler 1999].

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